

**MOUNT GELLIBRAND, Australia**WMO: **958450**Lat: **38.2333S**Lon: **143.7925E**Elev: **860**StdP: **14.25**Time Zone: **10.00 (AUS)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 37.9       | 39.2 | 31.8                          | 27.0 | 61.5 | 34.0 | 29.6 | 52.6 | 45.7                  | 48.5 | 42.4 | 49.5 | 15.7                  | 280  | 0.777 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 21.6                   | 92.5            | 63.8 | 87.2 | 62.8 | 82.2 | 62.0 | 67.6                | 79.1 | 65.9 | 77.5 | 64.5 | 76.3 | 19.3                 | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.0                            | 95.4 | 68.8 | 62.9 | 88.6 | 67.8 | 61.0 | 82.9 | 66.4 | 32.5          | 78.4 | 31.1 | 77.3 | 30.0 | 76.6 | 74.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 42.5          | 36.1 | 31.6 | 34.9                       | 102.4 | 1.2                | 4.3 | 34.0                                               | 105.5 | 33.3       | 108.0 | 32.6       | 110.4 | 31.7       | 113.5 |    |    |
| (5)               |      |      | 34.3                       | 71.3  | 1.3                | 2.1 | 33.4                                               | 72.8  | 32.6       | 74.0  | 31.9       | 75.2  | 30.9       | 76.7  |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 55.8   | 65.2  | 65.0  | 62.6  | 58.0  | 52.5  | 48.3  | 47.0  | 47.8  | 50.2  | 53.7  | 58.0  | 61.4  |  |  |
|                                                                        | DBStd       | 8.64   | 7.67  | 6.79  | 6.99  | 5.75  | 4.33  | 3.01  | 2.79  | 3.66  | 4.57  | 6.23  | 7.05  | 7.17  |  |  |
|                                                                        | HDD50       | 359    | 0     | 0     | 0     | 4     | 20    | 67    | 99    | 89    | 50    | 24    | 5     | 1     |  |  |
|                                                                        | HDD65       | 3703   | 91    | 77    | 133   | 224   | 388   | 500   | 557   | 533   | 444   | 357   | 238   | 161   |  |  |
|                                                                        | CDD50       | 2458   | 470   | 420   | 391   | 243   | 97    | 16    | 7     | 22    | 56    | 139   | 243   | 354   |  |  |
|                                                                        | CDD65       | 327    | 95    | 77    | 59    | 13    | 0     | 0     | 0     | 0     | 0     | 7     | 27    | 49    |  |  |
|                                                                        | CDH74       | 3545   | 1114  | 713   | 569   | 90    | 1     | 0     | 0     | 0     | 3     | 100   | 311   | 644   |  |  |
|                                                                        | CDH80       | 1523   | 560   | 304   | 225   | 16    | 0     | 0     | 0     | 0     | 0     | 24    | 99    | 295   |  |  |
| (14) Wind                                                              | WSAvg       | 15.9   | 15.2  | 15.0  | 15.1  | 14.6  | 16.0  | 15.1  | 17.7  | 17.4  | 16.7  | 16.6  | 15.5  | 15.6  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 20.5   | 1.5   | 1.3   | 1.2   | 1.5   | 1.5   | 2.1   | 2.0   | 2.0   | 2.4   | 2.0   | 1.7   | 1.4   |  |  |
|                                                                        | PrecMax     | 29.5   | 5.6   | 3.6   | 3.1   | 5.1   | 2.7   | 4.5   | 3.7   | 3.7   | 5.0   | 3.9   | 3.9   | 4.9   |  |  |
|                                                                        | PrecMin     | 11.8   | 0.0   | 0.0   | 0.4   | 0.1   | 0.3   | 0.5   | 0.9   | 0.6   | 1.0   | 0.2   | 0.4   | 0.0   |  |  |
|                                                                        | PrecStd     | 4.0    | 1.3   | 0.9   | 0.7   | 1.1   | 0.7   | 0.8   | 0.7   | 0.9   | 1.0   | 1.1   | 0.9   | 1.0   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 101.5 | 95.2  | 94.0  | 82.1  | 70.2  | 59.6  | 58.3  | 63.6  | 71.1  | 83.3  | 88.7  | 96.4  |  |  |
|                                                                        |             | MCWB   | 65.0  | 65.3  | 63.0  | 59.9  | 55.5  | 52.8  | 52.5  | 53.7  | 55.9  | 59.5  | 63.1  | 62.7  |  |  |
|                                                                        | 2%          | DB     | 93.0  | 88.5  | 86.3  | 76.0  | 65.1  | 56.6  | 55.3  | 59.6  | 65.7  | 75.8  | 82.5  | 88.7  |  |  |
|                                                                        |             | MCWB   | 63.8  | 63.5  | 61.9  | 57.6  | 54.7  | 52.0  | 50.9  | 51.5  | 55.6  | 58.4  | 61.9  | 62.2  |  |  |
|                                                                        | 5%          | DB     | 86.0  | 82.9  | 80.3  | 71.5  | 61.8  | 54.7  | 53.3  | 56.2  | 61.6  | 70.1  | 76.7  | 81.6  |  |  |
|                                                                        |             | MCWB   | 62.8  | 63.8  | 61.2  | 56.9  | 54.0  | 51.3  | 49.4  | 50.3  | 53.5  | 57.0  | 61.2  | 61.2  |  |  |
|                                                                        | 10%         | DB     | 79.0  | 77.0  | 74.4  | 67.1  | 59.0  | 53.0  | 51.7  | 53.6  | 58.3  | 64.5  | 70.9  | 74.8  |  |  |
|                                                                        |             | MCWB   | 62.3  | 62.7  | 60.6  | 56.5  | 53.5  | 50.2  | 48.6  | 49.1  | 51.8  | 55.4  | 59.1  | 60.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 70.2  | 69.7  | 67.2  | 62.9  | 59.9  | 55.4  | 54.9  | 55.6  | 59.7  | 63.6  | 68.1  | 68.6  |  |  |
|                                                                        |             | MCDB   | 83.8  | 82.5  | 77.0  | 72.5  | 62.4  | 56.9  | 56.2  | 61.5  | 65.9  | 74.6  | 79.4  | 76.3  |  |  |
|                                                                        | 2%          | WB     | 67.5  | 67.1  | 64.9  | 60.9  | 57.3  | 53.4  | 52.0  | 53.0  | 57.1  | 60.3  | 64.7  | 65.3  |  |  |
|                                                                        |             | MCDB   | 80.7  | 77.6  | 76.1  | 68.3  | 61.3  | 55.2  | 53.6  | 57.1  | 62.9  | 70.9  | 75.2  | 77.9  |  |  |
|                                                                        | 5%          | WB     | 65.6  | 65.5  | 63.4  | 59.4  | 55.6  | 51.9  | 50.4  | 51.2  | 54.8  | 58.2  | 62.5  | 63.3  |  |  |
|                                                                        |             | MCDB   | 79.3  | 75.7  | 74.1  | 65.9  | 60.0  | 53.8  | 52.1  | 54.6  | 60.4  | 67.4  | 72.9  | 76.3  |  |  |
|                                                                        | 10%         | WB     | 63.8  | 63.8  | 61.7  | 57.9  | 54.0  | 50.7  | 49.2  | 49.8  | 52.6  | 56.0  | 60.1  | 61.4  |  |  |
|                                                                        |             | MCDB   | 76.9  | 73.9  | 71.6  | 64.4  | 57.8  | 52.5  | 51.0  | 52.9  | 57.4  | 63.3  | 69.4  | 73.7  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 21.6  | 19.7  | 17.6  | 14.6  | 10.7  | 8.8   | 9.2   | 10.7  | 13.0  | 16.2  | 18.0  | 20.3  |  |  |
|                                                                        | 5% DB       | MCDBR  | 33.2  | 29.9  | 27.3  | 21.5  | 14.7  | 10.4  | 10.7  | 13.8  | 18.0  | 25.9  | 28.8  | 33.0  |  |  |
|                                                                        |             | MCWBR  | 8.8   | 8.4   | 8.0   | 7.4   | 7.9   | 7.2   | 7.5   | 8.3   | 10.5  | 11.6  | 11.2  | 9.8   |  |  |
|                                                                        | 5% WB       | MCDBR  | 25.9  | 23.8  | 21.9  | 17.0  | 12.2  | 9.2   | 9.8   | 12.2  | 16.2  | 21.7  | 23.4  | 24.8  |  |  |
|                                                                        |             | MCWBR  | 10.1  | 9.1   | 8.7   | 8.1   | 8.0   | 7.2   | 7.7   | 8.4   | 10.6  | 11.8  | 11.3  | 10.0  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.340 | 0.338 | 0.319 | 0.312 | 0.303 | 0.293 | 0.294 | 0.308 | 0.323 | 0.328 | 0.336 | 0.329 |  |  |
|                                                                        | taud        |        | 2.511 | 2.528 | 2.583 | 2.577 | 2.578 | 2.596 | 2.585 | 2.537 | 2.486 | 2.485 | 2.491 | 2.535 |  |  |
|                                                                        | Ebn at Noon |        | 315   | 308   | 301   | 284   | 266   | 260   | 267   | 281   | 295   | 307   | 314   | 320   |  |  |
|                                                                        | Edh at Noon |        | 35    | 33    | 29    | 25    | 22    | 20    | 22    | 26    | 31    | 34    | 36    | 35    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2245  | 1951  | 1525  | 1062  | 709   | 560   | 620   | 878   | 1253  | 1681  | 1971  | 2223  |  |  |
|                                                                        | RadStd      |        | 138   | 90    | 86    | 63    | 33    | 41    | 37    | 49    | 76    | 130   | 122   | 142   |  |  |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.06 | N/A     | N/A    | +2.93   | +1.75 | +2.34 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (36 neighbors) |  | +0.88 | N/A     | N/A    | +1.55   | N/A   | N/A   | -23         | -232  | +308  | +92   |

Nomenclature: See separate page